

Predicting Credit Risk Using Supervised Classification: A Cost-Sensitive Evaluation on the German Credit Dataset

Mohammed Adnan*

School of Computer Science

University of Nottingham

Nottingham, UK

Email: psxm4@nottingham.ac.uk

Abstract—This study investigates whether supervised machine learning can predict credit risk from loan applicant data and whether cost of misclassification changes which model a bank should prefer. Using the Statlog German Credit dataset [1] i trained and compared four classifiers - Logistic Regression, Decision Tree, Random Forest and K-Nearest Neighbours - under stratified 5-fold cross validation. Overall, The results show that Logistic Regression with threshold tuning at 0.25 gives best balance between catching bad applicants and minimising total cost, reducing cost from 151 to 95 which is a reduction of around 37%. And i also observe that optimising decision threshold based on asymmetric cost matrix is more impactful than choosing a more complex model. The full project pipeline is shown in Fig. 1.

Index Terms—Credit risk, supervised classification, logistic regression, decision tree, random forest, K-nearest neighbours, cost-sensitive evaluation, threshold tuning, stratified cross-validation, machine learning

I. INTRODUCTION

Credit risk assessment is one of the most crucial decisions a bank has to make. Approving a loan to someone who will default is significantly more costly than rejecting a good applicant and with that the cost of these two errors is not equal. The UCI German Credit dataset [1] formalises this through a official cost matrix where misclassifying a bad applicant as good costs 5 times more than reverse which signifies that standard accuracy is not a good metric for this particular problem.

Machine learning has been widely applied to credit risk classification [2], [3]. Studies have shown that ensemble methods like Random Forest generally outperform single classifiers on tabular financial data [4] but choice of evaluation metric and decision threshold can matter more than model complexity in cost-sensitive settings [5]. This study investigates whether that holds true on German Credit dataset and compares four classifiers under a cost-aware evaluation protocol. The research question is: can we predict whether a loan applicant is a good or bad credit risk and does cost of misclassification change which model a bank should prefer?

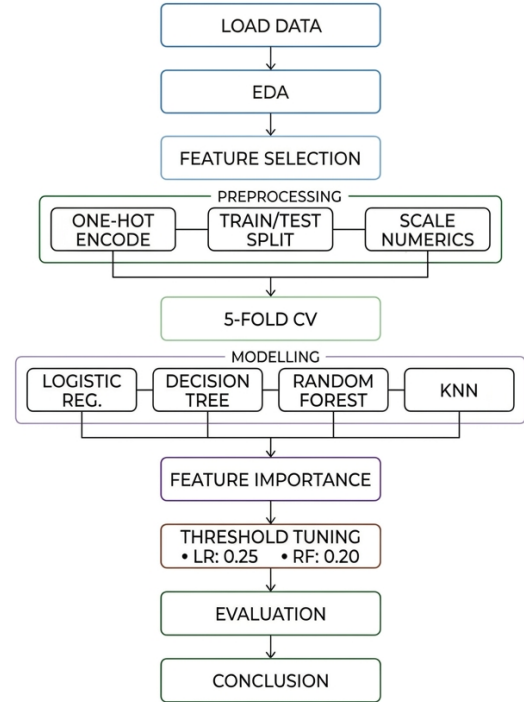


Fig. 1. Project pipeline diagram.

II. DATASET AND EXPLORATORY ANALYSIS

The Statlog German Credit dataset [1] was collected in Germany in year 1994 and contains 1000 loan applicants with 20 features - a mix of numerical and categorical attributes. The target variable indicates good credit risk (700 applicants, 70%) or bad credit risk (300 applicants, 30%) as shown in Fig. 2. This moderate class imbalance signifies accuracy alone would be misleading - confirmed by Dummy Classifier results in Section V.

Exploratory analysis revealed no missing values in dataset. Three features were removed before modelling - `own_telephone`, `num_dependents` and `residence_since` — as they showed near-zero class



Fig. 2. Class distribution bar chart.

separation variance of 0.0006, 0.13 and 1.22 respectively and no logical relevance to credit risk. This left 17 features which became 45 after one-hot encoding. I observe that `checking_status` and `credit_history` show highest bad rate variance across their categories which signifies these are most discriminative categorical features - later validated by feature importance analysis in Fig. 6.

III. RELATED WORKS

Baesens et al. [2] evaluated multiple classifiers on credit scoring datasets and found that simple models like Logistic Regression often perform competitively against complex ones - consistent with our results in Section V where LR outperforms Random Forest and KNN at default threshold. Breiman [4] introduced Random Forests as an ensemble method that reduces variance through bagging and generally outperforms single decision trees. Hand and Henley [3] argue that misclassification costs are asymmetric in lending and standard metrics like accuracy can be misleading. Threshold tuning as a method to align predictions with cost structures is discussed by Provost and Fawcett [5] through ROC analysis - directly supporting our approach in Section IV-C. The scikit-learn framework [6] is used for all modelling and Kohavi [7] provides justification for k-fold cross validation as a reliable evaluation protocol.

IV. METHODOLOGY

A. Preprocessing

Categorical features were one-hot encoded with `drop_first=True` to avoid dummy variable trap. Data was split 80/20 into train (800 samples) and test (200 samples) using stratified split to preserve 70/30 class ratio. `StandardScaler` was fitted only on training data and applied to test - a crucial leakage safeguard [6]. After encoding final feature count became 45.

B. Evaluation Protocol

Stratified 5-Fold Cross Validation was used on training data rather than a fixed validation set - a good practise for dataset of this size [7]. Each fold preserves 70/30 class ratio verified fold-by-fold. Primary metric for Grid Search is F1 on Bad class because of class imbalance, defined formally as:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

Where $\text{Precision} = \frac{TP}{TP+FP}$ and $\text{Recall} = \frac{TP}{TP+FN}$. Cost sensitive evaluation uses the UCI cost matrix - $FN \times 5$ and $FP \times 1$ - as it reflects real financial impact of each decision. The total cost is calculated as:

$$\text{Total Cost} = (FN \times 5) + (FP \times 1) \quad (2)$$

Where FN is number of bad applicants misclassified as good and FP is number of good applicants misclassified as bad. Final evaluation is on held-out test set never touched during training.

C. Models and Hyperparameter Tuning

Two baselines were established - DummyClassifier (always predicts Good) and Logistic Regression as ML baseline. Followed next with Decision Tree tuned over `max_depth` (3, 4, 5, 6, 7, 8, 10, None), Random Forest over `n_estimators` (100, 200) and `max_depth` (5, 8, None) and KNN over `n_neighbors` (3, 5, 7, 9, 11, 15, 21). All Grid Searches use Stratified 5-Fold CV with F1 as scoring metric. Threshold tuning was applied to LR and RF by sweeping decision boundary from 0.1 to 0.9 and selecting threshold that minimises total cost.

V. RESULTS

Table I shows final comparison of all models on test set.

TABLE I
FINAL MODEL COMPARISON. COST MATRIX: $FN \times 5 + FP \times 1$.

Model	Acc.	F1	Recall	Prec.	AUC	FN	FP	Cost
Dummy	0.700	0.000	0.000	0.000	0.500	60	0	300
LR	0.785	0.6055	0.5500	0.6735	0.8046	27	16	151
DT	0.610	0.4179	0.4667	0.3784	0.5690	32	46	206
RF	0.780	0.5510	0.4500	0.7105	0.7939	33	11	176
KNN	0.730	0.5091	0.4667	0.5600	0.6899	32	22	182
LR Tuned	0.705	0.6335	0.8500	0.5050	0.8046	9	50	95
RF Tuned	0.580	0.5800	0.9667	0.4143	0.7939	2	82	92

Overall, The Dummy Classifier confirms naive baseline - F1 0.00 and total cost of 300 as it misses all 60 bad applicants. All four real models beat this convincingly. Logistic Regression achieves best default performance with F1 0.6055, ROC-AUC 0.8046 and total cost 151 - correctly identifying 33 out of 60 bad applicants. CV results confirm this with mean F1 of 0.5042 ± 0.0988 and ROC-AUC of 0.7763 ± 0.0558 across 5 folds. Random Forest shows best precision of 0.7105 but lower recall of 0.45 - catching only 27 bad applicants with total cost 176 which signifies it is more conservative than LR. Decision Tree performs worst with F1 0.4179, ROC-AUC 0.5690 and total cost 206 - best `max_depth` from Grid Search was None with CV F1 0.4955 dropping to test F1 0.4179 which is a clear sign of overfitting. KNN achieves F1 0.5091, ROC-AUC 0.6899 and total cost 182 with best `n_neighbors` of 3.

As seen in Fig. 3 LR and RF show strong ROC curves with AUC of 0.8046 and 0.7939 respectively - well above random classifier diagonal. KNN falls to 0.6899 and Decision Tree to 0.5690 confirming their weaker discrimination ability.

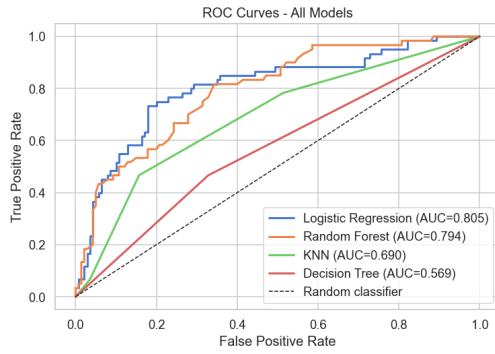


Fig. 3. ROC curves for all models.

TABLE II
COST REDUCTION AFTER THRESHOLD TUNING.

Model	Default Cost	Threshold	Tuned Cost	Reduction
Log. Reg.	151	0.25	95	37%
Rand. Forest	176	0.20	92	48%

Threshold tuning has a major impact on results. The cost reduction is shown in Table II below.

Lowering LR threshold from 0.5 to 0.25 drops total cost from 151 to 95 - recall improves from 0.55 to 0.85 meaning i now catch 51 out of 60 bad applicants instead of 33 and false negatives drop from 27 to only 9. For Random Forest lowering threshold to 0.20 drops cost from 176 to 92 - catching 58 out of 60 bad applicants with only 2 false negatives but false positives increase dramatically from 11 to 82. Threshold tuning results for LR and RF are shown in Fig. 4 and Fig. 5 respectively.

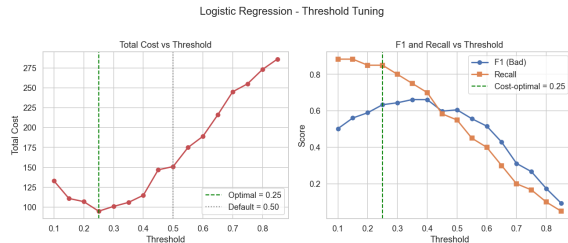


Fig. 4. LR threshold tuning plot.

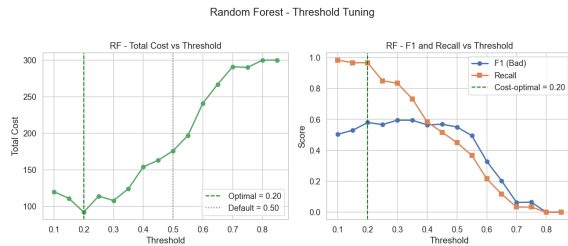


Fig. 5. RF threshold tuning plot.

Feature importance analysis from Random Forest is shown in Fig. 6. Top three features are all numerical

- credit_amount (0.1282), age (0.0980) and duration (0.0937) - followed by checking_status_A14 (0.0704) and installment_commitment (0.0422) which signifies loan characteristics matter more than personal demographics in this dataset.

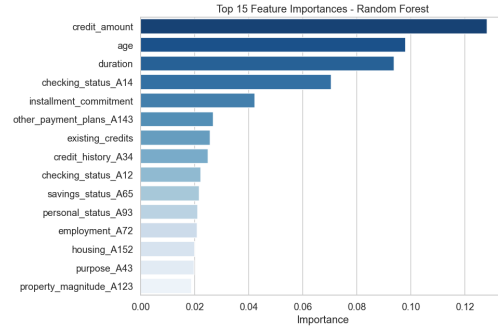


Fig. 6. Feature importance bar chart.

CV F1 stability is shown in Fig. 7. LR shows highest mean F1 of 0.5042 but also highest variance at ± 0.0988 . Followed by DT at 0.4955 ± 0.0723 , RF at 0.4488 ± 0.0826 and KNN at 0.4379 ± 0.0540 - KNN is most stable but consistently weakest.

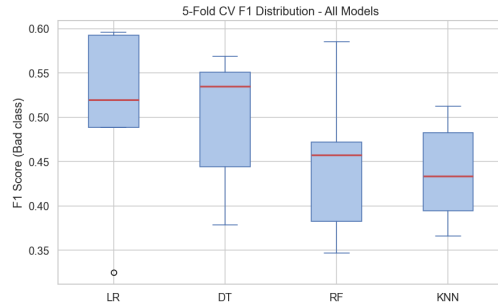


Fig. 7. CV F1 stability boxplot.

CV cost stability is shown in Fig. 8. DT shows lowest mean cost of 144.4 ± 21.1 and KNN is most stable at 166.0 ± 13.9 . LR shows 148.4 ± 26.1 and RF shows highest mean cost of 165.8 ± 20.5 . i observe that DT has lowest mean CV cost but worst test cost of 206 - a clear sign of overfitting that would not be visible from CV alone. This signifies cost-aware evaluation on held-out test set is crucial.

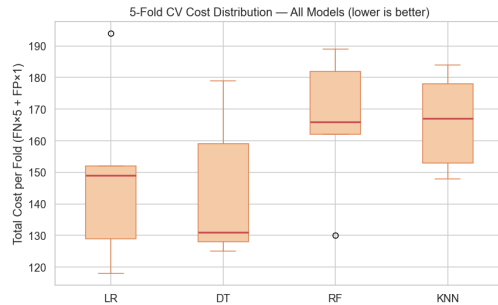


Fig. 8. CV cost stability boxplot.

VI. DISCUSSION

The results show that model complexity alone does not determine which model is best for credit risk. Logistic Regression - simplest of real models - outperforms Random Forest, Decision Tree and KNN on F1 and ROC-AUC at default threshold as shown in Table I. This is consistent with findings of Baesens et al. [2] who observed that simple models often perform competitively on credit scoring datasets.

The most significant finding is impact of threshold tuning shown in Fig. 4 and Fig. 5. By shifting decision boundary below 0.5 i align model predictions with asymmetric cost structure of problem as discussed by Provost and Fawcett [5]. With that a bank choosing between LR tuned (cost 95, FP 50) and RF tuned (cost 92, FP 82) would need to consider their risk appetite - RF tuned achieves slightly lower cost but rejects significantly more good applicants which could have a major impact on customer relationships.

Feature importance in Fig. 6 shows `credit_amount`, age and duration are strongest predictors which aligns with credit risk literature [3] where financial behaviour is generally a stronger predictor than demographic attributes. And i also observe from Fig. 7 and Fig. 8 that DT has lowest mean CV cost but worst test cost - a contradiction that highlights why relying only on CV metrics without test set evaluation can be misleading in a real lending situation.

VII. CONCLUSION

This study shows that supervised classification can predict credit risk with decent performance on German Credit dataset. Logistic Regression with threshold tuning at 0.25 gives best balance between catching bad applicants and minimising total financial cost as shown in Table I and Table II. And i observe that cost-sensitive evaluation and threshold tuning are more impactful than model complexity - a finding supported by both results and related literature [3], [5].

In todays world where financial institutions are increasingly using automated systems for credit decisions [3] it is crucial that evaluation goes beyond accuracy and considers real cost of errors. A simple adjustment to decision threshold reduces total cost by around 37% for LR and around 48% for RF without changing the underlying model at all. Future work could explore fairness analysis on `personal_status` feature which encodes gender and marital status and replication on more recent credit datasets to generalise conclusions beyond the 1994 German context.

REFERENCES

- [1] H. Hofmann, "Statlog (German Credit Data)," UCI Machine Learning Repository, 1994.
- [2] B. Baesens, T. Van Gestel, S. Viaene, M. Stepanova, J. Suykens and J. Vanthienen, "Benchmarking state-of-the-art classification algorithms for credit scoring," *Journal of the Operational Research Society*, vol. 54, no. 6, pp. 627–635, 2003.
- [3] D. J. Hand and W. E. Henley, "Statistical classification methods in consumer credit scoring: a review," *Journal of the Royal Statistical Society: Series A*, vol. 160, no. 3, pp. 523–541, 1997.
- [4] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [5] T. Provost and T. Fawcett, "Robust classification for imprecise environments," *Machine Learning*, vol. 42, no. 3, pp. 203–231, 2001.
- [6] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [7] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in *Proc. IJCAI*, 1995, pp. 1137–1143.